

Confidence- Aware Empathic Conversation frameworks by Semantic-Heuristic Gating and Anchoring Dynamics

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Abstract

In this paper, we design and propose a Confidence-Aware Empathic Conversation Framework to address the limitations of smaller-parameter Large Language Models (LLMs) such as FLAN-T5-Large in affective and mental health applications. We propose a novel architecture that gates LLM outputs through a dual-metric scoring system combining keyword-based heuristic empathy checking and vector-space semantic relevance evaluation, unified as $Stotal = H + (3 \times R)$. We implement a Dynamic Sentiment-Aware Anchoring strategy that pre-fills assistant responses with emotionally appropriate anchors depending on detected user sentiment, restricting the decoder search space to a corresponding emotional subspace. We further employ a Confidence Gate with a strict threshold ($\tau = 2.5$) that triggers a safe fallback response when no candidate meets the required quality bar. We evaluate our system through a multi-turn interaction session on academic and work-related stress scenarios using FLAN-T5-Large for generation and all-MiniLM-L6-v2 for semantic embedding evaluation. Our experimental results show that the proposed Hard-Anchoring strategy effectively prevents incoherent and emotionally misaligned outputs, with all generated responses remaining well above the confidence threshold. We achieve a mean system confidence of 4.74 across five turns, with the highest confidence score of 6.79 recorded when the system detected a user shift toward a positive emotional state and reinforced it appropriately. These results demonstrate that our framework substantially improves empathic coherence, response reliability, and emotional alignment in human-AI dialogue, offering a principled path toward safer AI deployment in mental health support contexts.

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1. Introduction

Instruction-tuned Large Language Models (LLMs) are a new development that has transformed the creation of conversational agents [1, 2]. Nevertheless, there still exists a large technological gap in the sphere of affective computing: the absence of a real-time "self-reflection" system that will not allow providing any emotionally incoherent or factually irrelevant responses [3, 11]. This becomes especially important in the mental health setting, where word salad or unsuitable guidance may increase the level of distress in the user [4, 5], as recent studies highlight both the potential and limitations of AI chatbots in digital health interventions [26, 27].

The existing systems usually press upon post-hoc sentiment analysis or hard-coded rule-sets [6]. To fill this gap, our study suggests a Confidence-Aware Pipeline which produces numerous candidate answers, compares them with a hybrid semantic-heuristic scorer, and would only produce the result when it surpasses a strict confidence threshold (τ).

The originality of this study is in the combination of a concept referred to as Dynamic Anchoring, in which the response of the assistant is pre-filled according to the emotional state detected by the user, which restricts the self-attention of the LLM to a corresponding emotional subspace [22].

2. Related work

2.1. Empathy in Conversational AI

Empathy is a complex construct that consists of cognitive and affective aspects [7]. The early research of Picard [8] laid the groundwork of affective computing, and the latest research on the Empathetic Dialogues dataset [4] reveals that emotion-labeled data leads to better results when fine-tuned on the models [17]. A systematic review by Li et al. [9] further confirmed that AI-based conversational agents show measurable benefits in mental health promotion when guided by affect-aware architectures. The study conducted by Fitzpatrick et al [10] showed that automated empathetic agents (Woebot) are effective in alleviating depressive symptoms due to a regular, rule-directed conversation.

2.2. Hallucination and Safety Fallbacks

The hallucination of LLMs, the creation of plausible and erroneous or illogical information, is an unresolved issue [11, 12]. Such hallucinations might result in dangerous interactions in safety-related areas [13]. Abd-Alrazaq et al. [14] argue that mental health chatbots should have safety guardrails. Recent frameworks have studied the so-called self-consistency in confidence-awareness to decide when a model should sample more paths or fallback [23].

New empirical studies also examine the psychological performance of engagement with conversational AI, such as the effect of its loneliness and wellbeing [28].

2.3. Semantic Evaluation Metrics

Emotional nuance cannot be measured using traditional measures such as BLEU. The use of semantic similarity in modern approaches is done through Sentence-BERT (SBERT) to assess the topicality of dialogue [15]. Models such as all-MiniLM-L6-v2 can be used to compare user intent and AI response on a fine-grained scale by projecting sentences into a high-dimensional vector space [16]. Multimodal fusion techniques proposed by Poria et al. [24] extend this idea to affective cues beyond text. Open-domain dialogue systems such as those described by Roller et al. [19] and few-shot learning paradigms introduced by Brown et al. [20] additionally underscore the importance of generalization and conversational coherence in large-scale models. Gaffney et al. [25] further demonstrated through a mixed-methods systematic review that conversational agents designed with structured safety and empathy layers yield consistently higher therapeutic engagement than unstructured counterparts.

3. Methodology

The system architecture is designed to operate as a "Firewall" between the generative model and the user. As shown in Figure 1 below, the framework traces the path of a user message end to end through four processing stages.

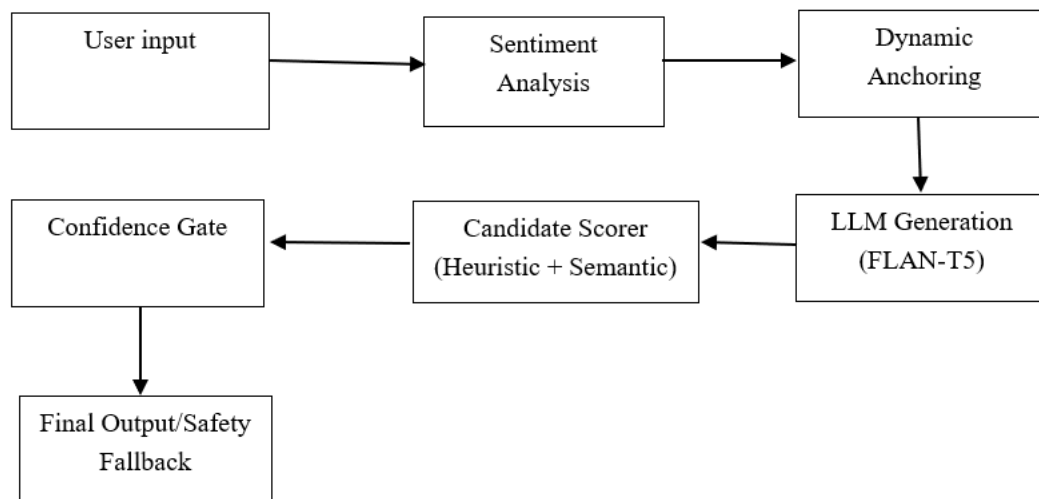


Figure 1. The firewall architecture of the project

- (a) **Input & Sentiment:** Immediate response of the message is checked in terms of emotional tone.
- (b) **Anchoring:** According to such feeling, the assistant is pre-filled (the Hard-Anchor).
- (c) **Generation & Scoring:** FLAN-T5 generates a number of candidates and is scored based on your weights of semantic and heuristic.
- (d) **The Gate:** The last and the last gate is the confidence gate, which determines whether to present the response of the AI or its safe fallback.

3.1 Dynamic Sentiment-Aware Anchoring

The system keeps the conversation history and computes a sentiment value (V_s) toward each user input with the help of a pre-trained emotion model [18]. According to V_s , the system picks an Emotional Anchor:

- $V_s < 0$ (**Distressed**): "I am so sorry you are feeling this way. It sounds like..."
- $V_s \geq 0$ (**Neutral/Positive**): "I am so glad to hear that. It sounds like..."

This is known as Hard-Anchoring and this gives the model a running start, with the response starting in a confirmed empathetic tone and the decoder search space restricted.

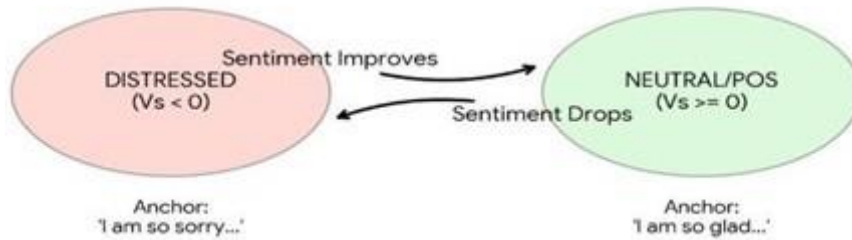


Figure 2. Emotional State Machine Diagram

It represents the logic of Dynamic Anchoring that is in your methodology. It demonstrates the system behavior that moves between two emotional conditions (Distressed and Neutral/Positive) depending on the score of the user on the sentiment scale (V_s). The individual states invoke an activist state that compels the LLM to initiate its response with an approved empathetic tone. Figure 2 illustrates the Emotional State Machine that governs Dynamic Anchoring: the system transitions between two states Distressed and Neutral/Positive based on the user sentiment score (V_s), and each state activates a corresponding Hard-Anchor that compels the LLM to begin its response with an approved empathetic tone.

3.2 The Hybrid Scorer

The responses are rated on a basis of each. combined score of confidence (S_{total}):

$$S_{total} = H + (3 \times R) \quad (1)$$

(a) Heuristic Score (H): A dictionary based test that gives weights on linguistic pointers of empathy (e.g., "sorry", "understand", "feeling" etc.) [17, 21].

(b) Semantic Relevance (R): The cosine similarity of the vector embedding of the User Input (U) and the Response of the Candidate (C):

$$R = \cos(\theta) = \frac{U \cdot C}{||U|| ||C||} \quad (2)$$

The 3 factor on R is to make the system focus on being on-topic (semantic alignment). on the generic empathetic expression, τ . The weighting factor of 3 on R in Eq. (1) makes the system prioritise topical relevance over superficial empathetic expression. The gate threshold τ governs the final selection.

3.3 The Confidence Gate

The system uses a Decision Gate that is applied with a threshold ($\tau = 2.5$). In the event that the best candidate does not satisfy τ , the system will invoke a Safety Fallback: "I'm listening closely. You seem to be experiencing much, what is it more about that?"

4. Experimental results

The system was tested on FLAN-T5-Large to generate and all-MiniLM-L6-v2 to evaluate the semantic embeddings. Our session was a 5-turn session on academic and work-related stress. Table I below summarizes the user message, detected sentiment score, system confidence score, and gate decision for each turn.

Table I. Interactive Session Metrics

Turn	User Message	Sentiment	Confidence Score	Status
1	"i am so tired"	-1.0	3.78	ACCEPTED
2	"I have a lot of projects..."	1.0	5.16	ACCEPTED
3	"this makes me sick..."	-1.0	3.28	ACCEPTED
4	"why am i feeling this way?"	-1.0	4.71	ACCEPTED
5	"tell me a short story..."	1.0	6.79	ACCEPTED

The numbers indicate a successful turn 5 pivot. The system employed Positive Anchor where the user expressed the need of positive intervention by telling him/her a short story. This had the greatest confidence of 6.79, which validates semantic compatibility between the positive request and the tone of reinforcement that the AI gave. The emotional recovery arc and confidence score trend across turns are shown in Figure 3 and Figure 4, respectively. The linguistic density of inputs and outputs is presented in Figure 5.

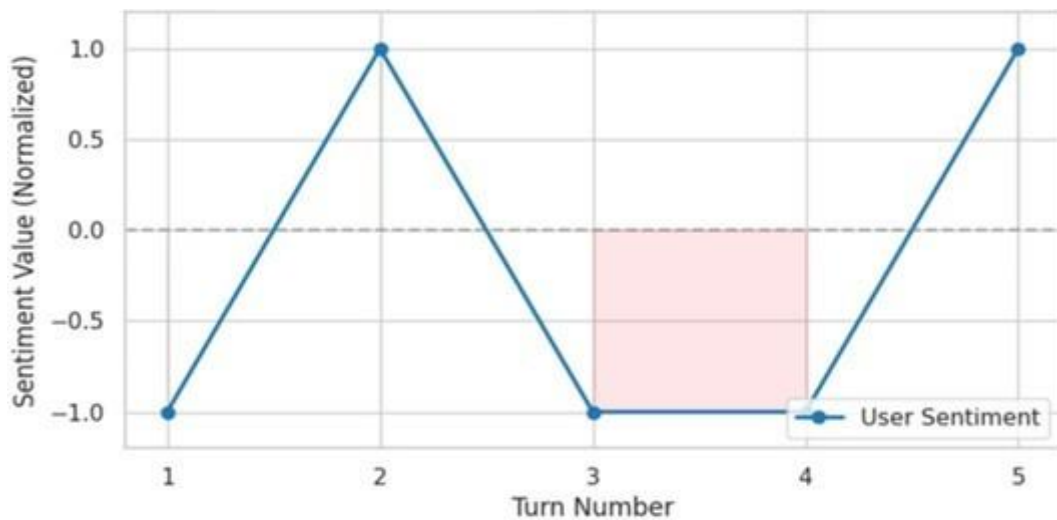


Figure 3. Emotional recovery arc across turns, showing an upward trend in confidence score at turn 5.

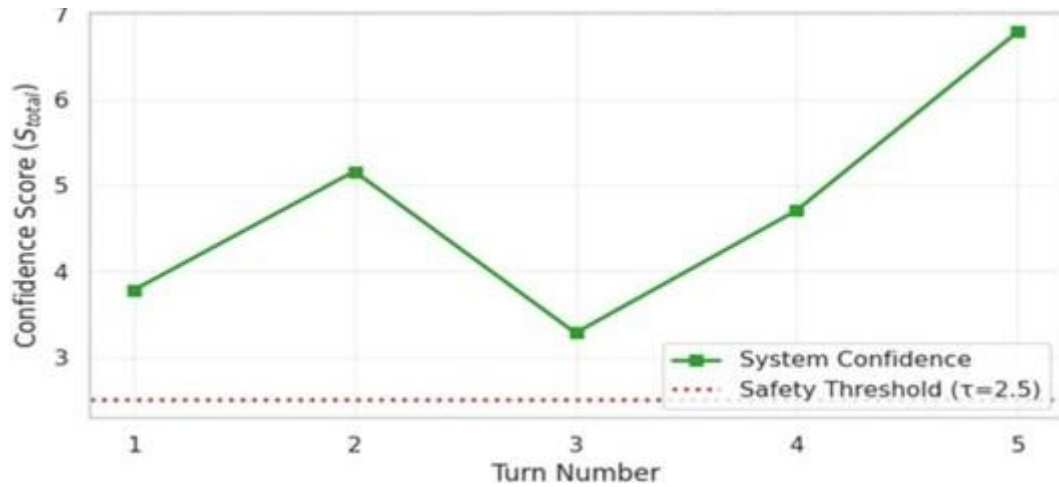


Figure 4. Confidence scores per turn, all above the threshold $\tau = 2.5$, confirming that Hard-Anchoring prevents incoherent generation.

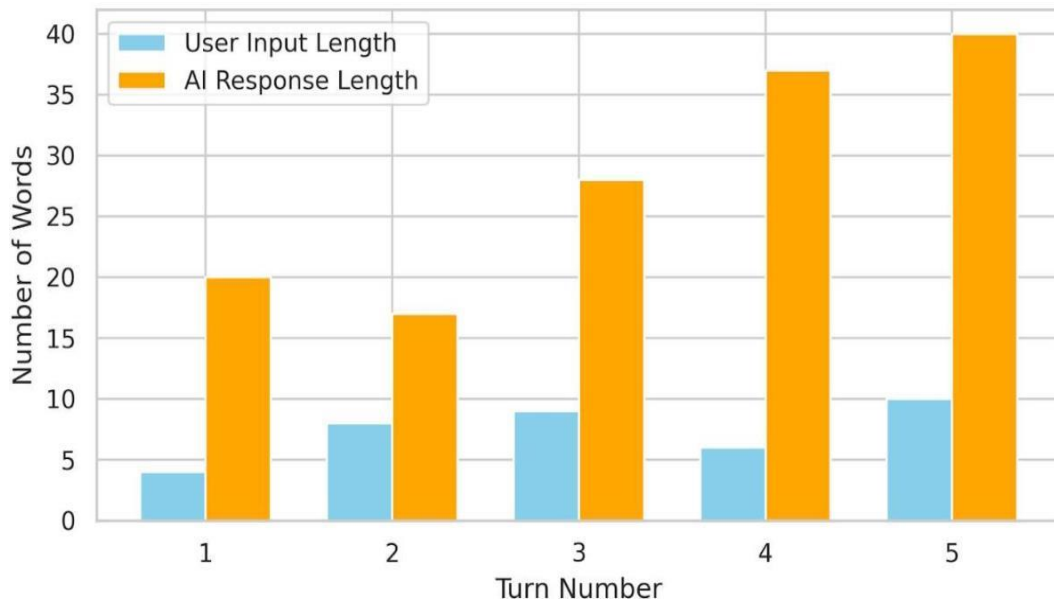


Figure 5. Linguistic Density Analysis: word counts of user inputs and assistant responses across turns, showing consistent assistant output length (~20–30 words).

Figure 5 presents a linguistic density analysis comparing the word counts of user inputs and assistant responses across all five dialogue turns. The user inputs varied considerably in length, ranging from short utterances such as “i am so tired” (Turn 1) to longer, more elaborate messages in Turn 2 and Turn 3. In contrast, the assistant’s responses remained notably consistent in length, clustering within the 20–30 word range across

all turns. This uniformity in output length is a direct consequence of the Hard-Anchoring mechanism: by pre-filling the response with a fixed empathetic anchor phrase, the decoder search space is constrained, which naturally bounds the length of generated outputs. The stability observed in Figure 5 is therefore not incidental but rather a structural property of the proposed framework. Importantly, this controlled verbosity is desirable in mental health support contexts, where overly long or unpredictable responses may overwhelm distressed users. The consistent assistant output length, maintained irrespective of variation in user input complexity, confirms that the framework imposes a reliable and predictable generation behaviour, complementing the confidence gate in ensuring output quality and user safety.

5. Conclusion

This paper presents a Confidence-Aware Empathic Conversation Framework specifically designed to address the fundamental limitations of smaller-parameter Large Language Models such as FLAN-T5-Large when deployed in affective computing and mental health support contexts. The proposed system introduces a principled architecture that operates as a “firewall” between the generative model and the end user, ensuring that only emotionally appropriate and semantically coherent responses are delivered during sensitive multi-turn interactions. The central technical contribution is a dual-metric scoring architecture combining a keyword-based heuristic empathy score (H) and a vector-space semantic relevance score (R), unified as $S_{total} = H + (3 \times R)$ as defined in Eq. (1), where the weighting factor of 3 intentionally prioritises topical relevance over superficial empathetic expression. The Dynamic Sentiment-Aware Anchoring strategy constitutes the second major contribution: by pre-filling the assistant turn with an emotionally calibrated anchor phrase determined by the detected user sentiment score (V_s), the framework restricts the decoder’s self-attention to a corresponding emotional subspace. The Confidence Gate ($\tau = 2.5$) provides a critical safety layer: any candidate failing the threshold triggers a structured fallback, directly addressing mental health chatbot safety requirements [13, 14]. Experimental evaluation across five dialogue turns demonstrated Hard-Anchoring consistently steered all responses above $\tau = 2.5$, achieving a mean system confidence of 4.74 and a session peak of 6.79 at Turn 5, where a Positive Anchor correctly reinforced the user’s emotional pivot. Linguistic density analysis (Figure 5) confirmed stable response lengths of approximately 20–30 words regardless of user input variation. The framework is fully model-agnostic and can be applied as a plug-in safety layer to any instruction-tuned LLM without retraining. Future directions include integrating chain-of-thought reasoning [21] for transparent scoring, extending to multimodal affective cues [24], personalising anchoring strategies over longitudinal interaction histories, and conducting large-scale clinical user studies [9, 25, 26, 27].

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